Glyph Mini 2040 MIDI Controller

Six keys, three knobs, an encoder, and a 128 x 64 OLED. The board sends USB MIDI; your computer or phone app makes the sound.

1. What you need

Glyph Mini 2040 and the six-switch PCB; six switches; three 10 kOhm linear pots; an encoder with push button; a 128 x 64 I2C OLED; and a USB data cable.

Download CircuitPython 10.x for the PCB Cupid Glyph Mini 2040 and the matching 10.x CircuitPython library bundle. The board setup was checked with CircuitPython 10.3.1.

2. Wiring

Part	GPIO	Also connect
OLED SDA / SCL	GP0 / GP1	VCC to 3V3; GND to GND
Switches 1-6	GP2 through GP7	Other leg of each to GND
Encoder A / B	GP8 / GP9	Common C to GND
Encoder button S2	GP10	S1 to GND
Pot A / B / C wipers	GP26 / GP27 / GP28	Outer legs to 3V3 and GND

Use 3.3 V, never 5 V. The encoder C and S1 share a PCB net. Check that this net reaches main GND; add a GND jumper if your board does not connect it.

3. Install CircuitPython and code

1. On Windows, Linux, or macOS, hold BOOT while connecting USB. Copy the board-specific CircuitPython 10.x .uf2 to RPI-RP2. The board should restart as CIRCUITPY.

2. Check CIRCUITPY/boot_out.txt for Board ID: pcbcupid_glyph_mini_2040.

3. Put these folders in CIRCUITPY/lib: adafruit_midi, adafruit_display_text, adafruit_display_shapes. Add these files: adafruit_displayio_ssd1306.mpy, adafruit_displayio_sh1106.mpy, adafruit_debouncer.mpy, adafruit_ticks.mpy, simpleio.mpy.

4. Copy glyph_mini_2040_midi_keyset/code.py to the root of CIRCUITPY as code.py. No main.bmp is needed. Later code changes require only replacing code.py, not reflashing the .uf2.

4. OLED choice

The code defaults to **SSD1306 at 0x3C**. For an SH1106, set OLED_DRIVER = "SH1106" near the top of code.py. If your 128 x 64 module uses address 0x3D, change OLED_ADDRESS to 0x3D. SDA is always GP0 and SCL is always GP1.

5. Make sound on Windows

Open VMPK's MIDI setup, choose the Glyph/CircuitPython USB MIDI device as **input**, keep a working synth as **output**, and enable MIDI Thru if offered. A mouse click making sound tests only the output. Press a physical switch to test the input.

6. Make sound on Linux

On Debian/Ubuntu, run **sudo apt install fluidsynth fluid-soundfont-gm alsa-utils**. Run **aconnect -l** to find the board's MIDI output; it may be named CircuitPython. Start the synth: **fluidsynth -a pulseaudio -m alsa_seq -o midi.autoconnect=1 /usr/share/sounds/sf2/FluidR3_GM.sf2**. Leave it running and press a key. See the README for other distributions and MIDI checks.

7. Make sound on macOS

Open GarageBand, create a Software Instrument track, choose a sound, and press a board key. If nothing plays, check for the Glyph/CircuitPython device in Audio MIDI Setup > Window > Show MIDI Studio; rescan if needed.

8. Controls

Control	Result
Six switches	Notes 61, 64, 66, 68, 73, 75; press = Note On, release = Note Off
Pot A / B / C	Modulation CC1 / pitch bend / expression CC11
Turn encoder	Move the OLED selection among three pots and six keys
Short press	Enter selected edit page; press again to leave
Long press in edit	Start or finish changing the selected value
Double press	All Notes Off on every MIDI channel

The startup music-note animation lasts about 1.7 seconds. Menu edits reset when the board restarts; change defaults in code.py to keep them.

9. First checks when something fails

What you see	Check
RPI-RP2, not CIRCUITPY	Release BOOT, tap RESET, then reinstall the correct .uf2 if needed.
CIRCUITPY but no MIDI	Read the serial console for a code.py error; verify the lib folders/files.
VMPK mouse works, board keys do not	Select the board as MIDI input; enable MIDI Thru; check serial key prints.
OLED blank or shifted	Check 3V3/GND, GP0/GP1, SSD1306 vs SH1106, and 0x3C vs 0x3D.
Encoder dead	Check GP8/GP9/GP10 and that common C/S1 reaches GND.
Knobs jitter	Check wipers at GP26-GP28, 3V3/GND, then try POT_DEADBAND = 3.

Full instructions, current code, and PCB link: github.com/pcbcupid/Glyph-MIDI-Controller
Original project documentation: @Dinesh v (2026). Updated for the six-key PCB and current firmware.